

The box is **drawn**, not written. Divide it however your prompt needs. If you need words, they go inside little boxes with arrows between them, in the order things happen. Sentences running across the page do not count.

**1****Membrane composition and fluidity**

COMP

**You should be able to:** Describe the fluid mosaic membrane and state how phospholipids, cholesterol, glycolipids, and integral and peripheral proteins contribute to its properties.

**PICK ONE AND DO THAT ONE**

- ☐ **A** Draw a patch of membrane in cross section, big enough to fill half your box. Put in phospholipids with heads and tails clearly different, cholesterol wedged between the tails, a transmembrane protein, a peripheral protein on the inner face, and a glycolipid with its sugar chain. Label which face is extracellular. One line: name the one component whose position alone tells you which side of the membrane you are looking at.

- ☐ **B** Draw the same membrane patch three times in a row: at normal body temperature, chilled, and warmed. In each, show how tightly the phospholipid tails pack and where cholesterol sits. Draw one tail with a double bond kink in it and one straight saturated tail. One line: cholesterol is called a fluidity buffer and not a stiffener, so say what the word buffer is doing there.

**DRAW IT****2****Determinants of permeability**

COMP

**You should be able to:** Rank molecules by their ability to cross a lipid bilayer unaided using size, charge, and lipid solubility, and predict which will require a protein.

**PICK ONE AND DO THAT ONE**

- ☐ **A** Draw a bilayer as a vertical wall down the middle of your page. Line up six molecules on the left: oxygen, carbon dioxide, water, a steroid hormone, glucose, and a sodium ion. Draw an arrow straight through the wall for each one that crosses unaided, a blocked arrow for each that cannot, and a protein set into the wall for each that needs help. One line: sodium is tiny and still cannot cross, so say what actually stopped it.

- ☐ **B** Draw a ranking ladder, most permeable at the top and least at the bottom, and place at least six molecules of your own choosing on it. Beside each rung write the properties that put it there: size, lipid solubility, and charge where it applies. One line: name the pair on your ladder that are close in size but far apart in rank, and say what separates them.

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**3**

COMP

**Signal types and range**

**You should be able to:** Classify a chemical signal as autocrine, paracrine, neurotransmitter, neurohormone, or hormone by its route and distance of travel.

**PICK ONE AND DO THAT ONE**

☐ **A** Draw one target cell in the middle of the box. Draw an arrow in from each of the five signal types, autocrine, paracrine, neurotransmitter, neurohormone, hormone. Label each arrow with how far it traveled and what carried it.

☐ **B** Make a five row table. Signal type, where it is released, how far it goes, what it travels through. Fill it from memory, then check one row against the book.

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**4**

COMP

**Receptor location and ligand solubility**

**You should be able to:** Predict whether a signal molecule binds a surface receptor or an intracellular receptor from its lipid solubility and relate that to the speed and duration of the response.

**PICK ONE AND DO THAT ONE**

☐ **A** Draw a cell with one surface receptor and one intracellular receptor. Show the path a lipid soluble signal takes and the path a water soluble one takes. Mark which response is faster and which lasts longer.

☐ **B** Write the two questions you would ask about any new signal molecule to predict where its receptor sits. Then answer both for a steroid and for a peptide hormone.

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**5****G protein coupled receptors**

COMP

**You should be able to:** Trace a G protein coupled receptor pathway from ligand binding through the G protein and amplifier enzyme to the second messenger and the cellular response.

**PICK ONE AND DO THAT ONE**

☐ **A** Draw the membrane with a receptor crossing it, and trace the pathway in numbered order: ligand binds, the G protein swaps GDP for GTP, the alpha subunit leaves, it switches on the amplifier enzyme, the enzyme makes second messenger, the messenger switches on a kinase, the kinase changes a target protein. Beside each step write roughly how many molecules that one step produced from one. One line: name the step where the biggest multiplication happened.

☐ **B** Draw a Gs pathway and a Gq pathway side by side, starting from the same membrane and ending at two different cellular responses. Label the amplifier enzyme and the second messenger on each. Then draw the off switch on both, GTP being hydrolyzed on the alpha subunit. One line: cholera toxin is that

**DRAW IT****6****Second messengers**

COMP

**You should be able to:** Identify cyclic AMP, IP3, diacylglycerol, and calcium as second messengers and state the enzyme that generates each and one target it activates.

**PICK ONE AND DO THAT ONE**

☐ **A** Draw a membrane and put all four second messengers on one page: cyclic AMP, IP3, diacylglycerol, and calcium. For each, draw where it comes from, the enzyme or channel that produced it, and an arrow to one thing it switches on. Keep DAG in the membrane and IP3 out in the cytosol, because that is where they actually go. One line: one of the four is not made by an enzyme at all, so name it and say where it comes from instead.

☐ **B** Draw phospholipase C cutting PIP2 in the membrane and follow both halves. Send IP3 to the endoplasmic reticulum and draw what comes out. Leave DAG where it is and draw what it activates. Show the two halves converging on one response. Draw the adenylyl cyclase and cyclic AMP route beside it for contrast. One line: cyclic AMP is destroyed by phosphodiesterase, so say what a drug

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**7****Catalytic receptors and intracellular receptors**

COMP

**You should be able to:** Compare a receptor enzyme such as the insulin receptor with an intracellular steroid receptor by mechanism and by time course of the response.

**PICK ONE AND DO THAT ONE**

☐ **A** Draw one cell with the insulin receptor in the membrane on one side and a steroid receptor sitting inside the cell on the other. Trace both from hormone arrival to final response, the steroid crossing the membrane and reaching DNA, the insulin receptor phosphorylating itself and moving a glucose transporter to the surface. Put a clock beside each showing how long the response takes. One line: name the one you could not use if you needed the effect ten seconds from now.

☐ **B** Draw one time axis with two response curves on it, a receptor enzyme response and a steroid response, each from hormone arrival through peak to fade. Make the onsets and durations honestly different. Under the axis, draw what is physically being changed in each case, an existing protein being modified or a new

**DRAW IT****8****Signal amplification**

COMP

**You should be able to:** Explain how a cascade amplifies a signal and estimate the size of the amplification across the steps of a given pathway.

**PICK ONE AND DO THAT ONE**

☐ **A** Draw a three step cascade as a widening funnel. Put a number on each step and multiply them down the side to get the total amplification.

☐ **B** Start from one ligand and write the count at each step of a cascade until you reach the response. One line per step, numbers only, no sentences.

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**9**

COMP

**Receptor modulation**

**You should be able to:** Define agonist, antagonist, competitive inhibition, up regulation, and down regulation and predict the effect of chronic ligand excess on target cell sensitivity.

**PICK ONE AND DO THAT ONE**

- ☐ **A** Draw one target cell three times. Normal, after a long stretch of scarce signal, after a long stretch of excess signal. Show the receptor count changing in each and label which is up regulation and which is down.
- ☐ **B** Define agonist, antagonist, up regulation and down regulation in your own words, one line each. Then explain in two lines why a patient who needs more opioid for the same effect is usually showing down regulation rather than drug seeking.

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**10**

COMP

**Signal termination**

**You should be able to:** Name three mechanisms that end a chemical signal and explain why a signal that cannot be terminated produces pathology.

**PICK ONE AND DO THAT ONE**

- ☐ **A** Draw one signal running from receptor to response across a cell, then add three different off switches to that same drawing at three different points: the ligand removed or destroyed outside the cell, the second messenger broken down inside, and the response protein having its phosphate taken off. Name the enzyme doing each. One line: pick the switch nearest the response and say what still goes wrong if only that one fails.
- ☐ **B** Pick a signal, not insulin, and draw it working normally, then draw it again with its termination broken. In the broken version show what piles up and where. Underneath, draw the effect on the tissue as two panels, one over minutes and one over weeks. One line: a signal that never shuts off eventually produces less response rather than more, so write what the cell did to itself.

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**11****Dose response relationships**

COMP

**You should be able to:** Plot a dose response curve, identify threshold, maximal response, and EC50, and compare a full agonist with a partial agonist on the same axes.

**PICK ONE AND DO THAT ONE**

☐ **A** Draw a dose response curve with log dose on the x axis and response as a percentage of maximum on the y. Mark threshold, EC50, and the maximal response, with construction lines dropped to the axes. Then plot a partial agonist on the same axes. One line: the partial agonist never reaches the top no matter how much you add, so say what that tells you about the drug that raising the dose cannot fix.

☐ **B** Draw three curves on one set of axes: a full agonist alone, the same agonist with a competitive antagonist added, and the same agonist with an antagonist that will not come off the receptor. Show which curve slides sideways and which one gets shorter. One line: only one of the two antagonists can be beaten by giving more agonist, so name it and say what feature of the graph proves it.

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